

Bank Marketing Analytics: Term Deposit Campaign Performance Analysis

1. Project Overview

This project analyses bank marketing transaction and campaign data across client demographics, operational contact metrics, and personal financial statuses. The goal is to identify customer behaviour trends, evaluate the factors influencing successful term deposit subscriptions, and deliver actionable insights for future marketing channel optimization.

2. Tools Used

- Python
- Pandas (Data manipulation and structured tables)
- NumPy (Numerical data structures)
- Matplotlib (Static data visualizations)
- Seaborn (Statistical grid plotting)
- Jupyter Notebook (Interactive code execution and environment)
- GitHub (Version control and project hosting)

3. Dataset Summary

- Source: UCI Machine Learning Repository (Bank Marketing Dataset)
- Scope: 45,211 rows and 17 core feature attributes
- Main Target Variable: 'y' (Has the client subscribed to a term deposit? Yes/No)

4. Key Focus Areas

- Demographics: Analysis of subscription variance across Job categories, Education, Age Groups, and Marital profiles.
- Touchpoint Metrics: Evaluating contact type efficiency and tracing the massive correlation impact of phone call duration.
- Financial Stability: Boxplot verification of customer average annual balances relative to conversion outcomes.